THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Materials Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student places a stone, a cork, an iron nail and a leaf in water. Which pair floats? 1

(a) cork and leaf **(b)** nail and stone **(c)** leaf and nail **(d)** stone and cork

2. Why does a stone sink in water but a boat made of iron float? 1

(a) water pushes iron **(b)** boats have no weight
(c) iron is lighter than stone **(d)** the boat's shape holds a lot of air

3. Why are cooking utensils such as pans often coated with a non-stick layer? 1

(a) to make them transparent **(b)** to make them heavier
(c) food does not stick to the surface **(d)** to make them dissolve

4. Which of the following pairs includes only soluble substances? 1

(a) sand and salt **(b)** salt and sugar **(c)** sand and chalk **(d)** chalk and sugar

5. Which test would show that a metal rod is a good conductor of heat? 1

(a) it breaks
(b) wax melts quickly along the rod when one end is heated
(c) it floats
(d) it dissolves

6. Why are clothes for winter often made of wool? 1

(a) wool dissolves **(b)** wool is a poor conductor of heat
(c) wool is shiny **(d)** wool is hard

7. A material is hard, shiny and a good conductor of heat. It is probably a: 1

(a) plastic **(b)** metal **(c)** paper **(d)** wood

8. Which of the following should be used to hold a hot kettle handle? 1

(a) an iron rod **(b)** a copper rod
(c) a steel handle **(d)** a thick wooden handle

9. Which of the following would you choose for making a water bottle? 1

(a) paper **(b)** steel **(c)** cotton **(d)** cloth

10. Which of the following would be a suitable material for making a window curtain which lets in some light? 1

(a) thin cotton cloth **(b)** wood
(c) thick metal sheet **(d)** brick

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do boats made of iron float while iron nails sink? 2

12. Why is a transparent glass not used for bathroom windows? 2

13. A student drops a sugar cube and a salt crystal in water. What is the observation? 2

14. Why is steel preferred over iron for making bridges and cars? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. You are given a mixture of sand, iron filings and sugar. Describe how to separate them using properties. 3

16. Give reasons: 3
(a) a doormat is rough;
(b) a tabletop is smooth;
(c) a window is transparent.

17. Why does a small stone tied to a large piece of thermocol float in water? 3

OR

Explain how to find which material is the best conductor of heat among iron, copper and aluminium rods. 3

18. How can one prove that a cork floats because it is lighter than water and not because it is hollow? 3

OR

Why are electrical appliances' pins made of metals and body of plastic? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Plan an investigation to find the best material for making a thermos flask lid among wood, plastic, steel and cork. 5

OR

Describe how to make a simple filter to separate a mixture of sand and salt in water. 5

20. Explain with experiments how the properties of materials (conduction, solubility, transparency, floating, hardness) can be tested. 5

OR

How are materials classified according to the sources? Give examples of animal, plant and mineral materials. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND